

Data Valuation for Machine Learning: Experiments and Analyses [Experiment, Analysis & Benchmark]

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Abstract

Data valuation assigns a quantitative measure of importance to data with respect to a given objective. In machine learning, the objective is often model performance, but could more broadly be any measurable property of the learning pipeline. It enables a wide range of uses including efficient training set selection, informed data curation, robustness and fairness in ML, and the design of incentive-aligned data markets. This paper conducts the first comprehensive experimental evaluation of data valuation methods spanning different granularities of data value. First, we propose a unified formal framework that introduces two complementary notions of data value: (i) coarse-grained valuation, which assigns a scalar value to a dataset as a whole, and (ii) fine-grained valuation, which assigns a scalar value to subsets of the data. Two generic operators, data value aggregation and data value attribution, formalize the relationship between the two granularities. We show that all existing state-of-the-art methods are recovered as instantiations of this framework. Second, we propose a taxonomy that classifies data valuation methods by the information they require from the learning pipeline. Third, we perform an extensive experimental evaluation of the efficiency/accuracy/memory tradeoffs of state-of-the-art valuation methods across eleven datasets. We describe the strengths and limitations of the methods, share new insights about their inner workings, and pinpoint promising research directions in the field.

CCS Concepts

• Information systems → Data management systems.

Keywords

Data Valuation, Economics, Machine Learning

1 Introduction

Motivation. Machine learning (ML) models derive their capabilities from the statistical properties of their training data, making data quality a critical factor in model generalization, robustness, and fairness [24, 50, 54, 58]. In practice, datasets often contain noise [21], shift [56], redundancy [80], and bias [49, 68], yet most ML pipelines treat all data samples equally. Data valuation addresses this problem by quantifying the contribution of data, from entire datasets to data units (i.e., non-empty subsets ranging from individual points to arbitrary

groups) to a given objective, which is typically model performance but could more broadly encompass any measurable property of the learning pipeline such as privacy or fairness. This helps practitioners to better understand model behavior [29, 33, 42, 51], and make decisions about data pricing in data marketplaces [2, 9, 53, 81], incentives in collaborative and federated learning [22, 66, 73], domain adaptation [23, 42, 78], data summarization [23, 38, 72, 75] or data curation [23, 37, 38, 42, 45, 74]. A compelling illustration of how data valuation can drive principled data curation arises in cancer diagnosis. AMONuSeg [79], the first public annotated African dataset for cancer cell analysis, required a full curation pipeline: source selection (identifying clinical sites across Morocco and Nigeria), data collection (acquiring slides under low-resource microscopy conditions), quality control (removing blurry and corrupted patches), and finally annotation (non-expert annotators tracing individual nucleus boundaries, with expert pathologists curating and validating labels through multiple consensus rounds spanning several months). Budget and time constraints left many valuable images annotated, highlighting the need for principled prioritization strategies as the dataset grows. Data valuation offers a path forward for extending the dataset. At the source selection and collection stages, coarse-grained valuation would rank candidate clinical sites by their distributional diversity, entirely label-free. Images from selected sites would then be annotated by non-experts. Fine-grained valuation would subsequently identify the highest-value instances to escalate for expert curation, ensuring that scarce pathologist time is spent where it matters most. Besides, data valuation has the potential to impact judicial settlements involving disputes over data, e.g., the copyright infringement lawsuit against OpenAI and Microsoft brought by the New York Times [67]. As ML models scale and data becomes more costly and regulated, principled reasoning about the value of data becomes essential for efficient, reliable and incentive-aligned systems.

Data Valuation Methods. Many data valuation methods have been proposed by different communities, including data management [36], ML [23, 38, 45, 47, 74, 75], statistics [5, 76] and economics [2, 19, 53]. Earlier works include Data Shapley [23], which exploits game theory to provide a theoretically grounded foundation for fair data valuation, and subsequent studies that improve its quality or efficiency [36, 44, 45, 47, 60, 74, 77]. Leave-One-Out (LOO) [11] quantifies the

contribution of data by measuring the drop in model performance when it is removed from the full training set and the model is retrained. However, LOO and game-theoretic approaches lack scalability due to the retraining cost, which can reach exponential complexity. Thus, researchers proposed gradient-based approaches, such as Influence Functions [42], which approximates LOO by replacing retraining with an analytical estimation. Other works focus on optimizing Influence Functions [10, 27, 28, 59], proposing new ways to study training trajectories [6, 55, 71], and using methods to approximate deep neural network behavior like the Neural Tangent Kernel (NTK) [51, 75]. Although these recent methods do not require retraining, they typically rely on the computation of the Hessian matrix, which is intractable for complex models [42]. Other works tackle the scalability challenge by proposing approaches that use intrinsic properties of the data, such as distance or volume, as a proxy for model performance [5, 36, 38, 76], though ignoring model-specific effects.

Challenges. Despite the growing body of work on data valuation, several challenges remain unaddressed. (1) *Scalability*: most data valuation methods do not scale to large datasets as they require repeated model retraining, which becomes prohibitive at scale. (2) *Validation set dependence*: training-based methods require access to a validation set, yet there is no consensus on whether this is a practical assumption, nor on how to select it and ensure its quality. (3) *Data heterogeneity*: most methods assume datasets share the same modality, schema, or structure, whereas real-world datasets are often heterogeneous. (4) *Privacy*: most methods assume access to the full dataset, while data sources may only expose a small sample. (5) *Versatility*: data value is model-specific, the same data may be valuable for one model but not another, and valuation becomes increasingly challenging for complex models such as neural networks and LLMs. (6) *Conflicting terminology*: data valuation has been studied across data management, economics, and statistics leading to inconsistent use of terminology that causes confusion and hinders cross-community progress [20]. (7) *Lack of comprehensive evaluation*: no study has systematically compared State-of-the-Art (SotA) data valuation methods under a common experimental framework covering different facets of the problem and accounting for scalability.

Contributions. We make the following contributions:

- We propose a formal framework that unifies existing data valuation methods, distinguishing between coarse-grained and fine-grained valuation with respect to a generic context C , and introducing two operators, data value aggregation, and data value attribution, that formalize the relationship between the two granularities. We show that all existing methods are recovered as instantiations of this framework.
- We propose a taxonomy that classifies data valuation methods by the information they require from the learning pipeline, separating task-independent from task-dependent methods, and further categorizing task-dependent methods as model-independent or model-dependent, with the latter further subdivided by the level of model access: black-box or white-box.
- We conduct the first comprehensive experimental evaluation for data valuation methods that covers the different problem

variants, performs a scalability study, and fine-tunes the methods to their efficiency/accuracy Pareto-frontiers.

- We evaluate the efficiency/accuracy/memory tradeoffs of fourteen SotA data valuation methods, across eleven datasets from various domains, including medicine, social science, gaming, biology and physics.

- We discuss the strengths and weaknesses of existing works, sharing new insights about their inner workings, and identify promising future research directions.

- We share all artifacts [12] to ensure reproducibility.

Our study fills a clear gap in the literature not addressed by prior works. Existing surveys [53, 63, 81] primarily focus on cooperative game-theoretic approaches. A recent survey [16] provides a high-level description of data valuation methods across different domains. None of these studies includes experiments. The OpenDataVal [37] benchmark is limited to fine-grained valuation, does not study scalability, covers only small datasets (up to 1K points vs. 7M in our paper), and does not perform a thorough parameter tuning. None of these works proposes a unified formal framework that reconciles the diverse terminology used across communities, covers fine- and coarse-grained valuation, or systematically classifies methods by the information they require from the learning pipeline.

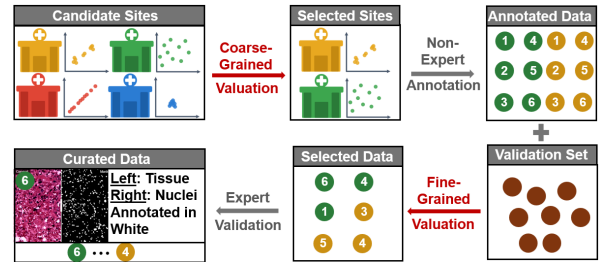


Figure 1: Data valuation for data curation

2 Preliminaries

This section introduces the key definitions and notation used throughout the paper. We first define the core objects of data valuation, datasets, data units, and context, and then introduce two complementary notions of data value: coarse-grained valuation, which assigns a scalar value to a dataset as a whole, and fine-grained valuation, which assigns a scalar value to individual data units. We further introduce two generic operators, data value aggregation and data value attribution, that formalize the relationship between the two granularities and unify existing data valuation methods under a single framework. In section 3, we present in detail how existing methods are recovered as special cases of this framework.

Definition 1 (Dataset). The *data domain* is a set $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} \subseteq \mathbb{R}^d$ is the *feature space* and $\mathcal{Y}$ is the *label space*. The integer d is the *dimensionality* of the data. The label space is generic: in classification, $\mathcal{Y}$ is finite; in regression, $\mathcal{Y} = \mathbb{R}$; and in unsupervised learning one may take $\mathcal{Y} = \{*\}$ as a singleton dummy label space.

A *data point* is an element $z \in \mathcal{Z}$. A *dataset* of size n is an finite indexed family of data points

$$\mathcal{D} = (z_i)_{i \in [n]} \in \mathcal{Z}^n,$$

where, repeated entries are allowed, i.e., it is possible that $z_i = z_j$ for distinct indices $i \neq j$.

For a dataset $\mathcal{D} \in \mathcal{Z}^n$, we denote by $X = (x_1, \dots, x_n) \in \mathbb{R}^{n \times d}$ and $Y = (y_1, \dots, y_n) \in \mathcal{Y}^n$ its *feature matrix* and *label vector*, respectively. Throughout the paper $\mathcal{Z}^*$ denotes the collection of datasets of all sizes, defined as

$$\mathcal{Z}^* := \bigcup_{n \geq 0} \mathcal{Z}^n.$$

Definition 2 (Data Unit). Let $\mathcal{D} = (z_i)_{i \in [n]}$ be a dataset. A *data unit* U is an indexed subfamily of $\mathcal{D}$ defined as

$$U = \mathcal{D}_I := (z_j)_{j \in I}$$

for some non-empty index set $I \subseteq [n]$.

We say that a family of data units $\mathcal{U} = (U_i)_{i \in [m]}$ is *covering* if, writing I_i the index set of U_i for each $i \in [m]$, we have

$$\bigcup_{i=1}^m I_i = [n].$$

The family $\mathcal{U}$ is called *pointwise* if $|I_i| = 1$ for each $i \in [m]$, and *groupwise* otherwise.

Remark 1. Throughout the paper, all set-theoretic operations on data unit objects are understood as the induced operations on their associated index sets. More precisely, if $U = \mathcal{D}_I$ and $W = \mathcal{D}_J$ for some index sets I and J , then

$$U \cup W := \mathcal{D}_{I \cup J}, \quad U \cap W := \mathcal{D}_{I \cap J}, \quad U \setminus W := \mathcal{D}_{I \setminus J}.$$

Generalizations to multiple data units are analogous.

Definition 3 (Context). A *context* $C \subseteq \mathcal{C}$ is the *contextual* information supplied to and used by a valuation method, beyond the dataset being valued. The set $\mathcal{C}$ denotes the collection of all admissible information including, for example, external datasets (e.g., a validation set $\mathcal{D}_{\text{val}}$), target distributions, learning tasks, or level of model access (e.g., black- or white-box).

Definition 4 (Data Valuation). A *data valuation* is a function $v : \mathcal{Z}^* \times \mathcal{C} \rightarrow \mathbb{R}$ that assigns a scalar value $v(\mathcal{D} \mid C) \in \mathbb{R}$ to any finite dataset $\mathcal{D} \in \mathcal{Z}^*$, given a context $C \subseteq \mathcal{C}$.

When the valuation method is applied directly to the whole dataset $\mathcal{D}$, we refer to it as a *coarse-grained valuation method*; such methods are surveyed in Section 3.2. In many situations, however, one is interested not only in the value of the full dataset, but also in quantifying the contribution of individual data units within it. This is captured by *attribution operators*, introduced next.

Definition 5 (Value Attribution). Let $\mathcal{D}$ be a dataset and $\mathcal{U} = (U_i)_{i \in [m]}$ be a family of data units of $\mathcal{D}$. A *value attribution operator* is a function ϕ that, given a data valuation v and a context¹ $C \subseteq \mathcal{C}$, assigns to each unit U_i a real number

$$\phi_i(\mathcal{D}, \mathcal{U} \mid v, C) \in \mathbb{R}, \quad i \in [m].$$

¹Although the data valuation v could technically be treated as part of the context, we separate it explicitly to highlight the attribution operator's dependence on it.

We simply write ϕ_i when there is no ambiguity and refer to it as the *fine-grained value* of U_i induced by (ϕ, v, C) . Data valuation methods based on attribution operators are called *fine-grained valuation methods*.

Example. A trivial attribution operator is obtained by applying the coarse-grained valuation independently to each data unit, namely

$$\phi_i^{\text{triv}}(\mathcal{D}, \mathcal{U} \mid v, C) = v(U_i \mid C), \quad i \in [m].$$

More sophisticated attribution operators, surveyed in Section 3.3 account for interactions among data units. Common constructions include:

- **Marginal contribution** (used by Leave-One-Out, Shapley, Banzhaf): ϕ_i is built from differences $v(\mathcal{D}_{S \cup \{i\}} \mid C) - v(\mathcal{D}_S \mid C)$ over subsets S of the remaining units, with method-specific weights.
- **Perturbation** (used by Data-OOB, Influence Subsample): ϕ_i compares the value of $\mathcal{D}$ with and without U_i across randomized subsets or bootstrap samples.
- **Differentiation** (used by OT Gradient): ϕ_i is a derivative of v with respect to the presence or weight of U_i .

Definition 6 (Data Value Aggregation). Let $\mathcal{D}$ be a dataset, let $\mathcal{U} = (U_i)_{i \in [m]}$ be a covering family of data units of $\mathcal{D}$, let $C \subseteq \mathcal{C}$ be a context, and let v be a data valuation. A *data value aggregation operator* is a rule α that maps a vector of fine-grained values $f = (f_j)_{j \in [m]} \in \mathbb{R}^m$ to a scalar

$$\alpha(\mathcal{D}, \mathcal{U}, C, v, f) \in \mathbb{R}.$$

Common choices include:

- **Summation:** $\alpha_\Sigma(\mathcal{D}, \mathcal{U}, C, v, f) := \sum_{i=1}^m f_i$.
- **Weighted Summation:** $\alpha_W(\mathcal{D}, \mathcal{U}, C, v, f) := \sum_{i=1}^m w_i f_i$. For instance, for units with overlaps, weights can be chosen to reflect multiplicities:

$$w_i = \frac{1}{\mu_i}, \quad \mu_i := \max_{k \in I_i} |\{j : k \in I_j\}|.$$

Summation and averaging are v -independent; more general aggregation rules may depend on v (e.g., weighting units by a v -derived importance). The signature $\alpha(\mathcal{D}, \mathcal{U}, C, v, f)$ accommodates both cases.

The following definition formalizes when attribution and aggregation operators act as “inverses” of each other.

Definition 7 (Consistency of a Valuation Scheme). A triple (v, ϕ, α) consisting of a data valuation v , an attribution operator ϕ , and an aggregation operator α is called *consistent* on a class $\mathcal{U}$ of covering families if, for every dataset $\mathcal{D}$, every $\mathcal{U} \in \mathcal{U}$, and every context $C \subseteq \mathcal{C}$,

$$v(\mathcal{D} \mid C) = \alpha(\mathcal{D}, \mathcal{U}, C, v, \phi(\mathcal{D}, \mathcal{U}, C, v)),$$

where $\phi(\mathcal{D}, \mathcal{U}, C, v) := (\phi_i)_{i \in [m]} \in \mathbb{R}^m$. When $\mathcal{U}$ is the class of all pointwise covering families, the scheme is called *pointwise-consistent*; when $\mathcal{U}$ admits groupwise coverings with possibly overlapping index sets, it is called *group-consistent*.

Consistency is analogous to Shapley’s *efficiency axiom*, generalized to arbitrary attribution/aggregation pairs. It is a structural property of valuation schemes (v, ϕ, α) that, when specialized to $\phi = \phi^{\text{Shap}}$ and $\alpha = \alpha_{\Sigma}$, recovers Shapley’s efficiency axiom. More generally, it captures the requirement that, for the chosen attribution operator ϕ and aggregation operator α , the fine-grained values produced by ϕ aggregate under α back to the coarse-grained value $v(\mathcal{D} \mid C)$.

3 Survey

In this section, we propose a taxonomy of data valuations, and briefly survey existing coarse- and fine-grained methods. Rather than aiming for exhaustive coverage, we focus on methods that are representative of the key methodological families.

3.1 Taxonomy

Our taxonomy (Fig. 2) classifies data valuation methods by the information they require from the learning pipeline. At the highest level, we distinguish *task-independent* and *task-dependent* methods, both requiring access to the target dataset $\mathcal{D}$. Task-independent methods rely solely on $\mathcal{D}$ ’s properties, such as its geometry or distribution, potentially measured relative to a reference distribution, and do not require access to a specific task or model. In contrast, task-dependent methods define data value specific to a given task $\mathcal{T}$ and are further categorized by their level of access to the target model $\mathcal{M}$. *Model-independent* methods exploit the task $\mathcal{T}$ (e.g., through its validation set $\mathcal{D}_{\text{val}}$) without requiring $\mathcal{M}$. *Model-dependent* methods either require *black-box* access to $\mathcal{M}$, typically estimating data value from the performance of $\mathcal{M}$ on random samples or bootstraps from $\mathcal{D}$, or *white-box* access to internal information from $\mathcal{M}$, such as gradients or parameters. Both white-box and black-box methods typically require a fixed hold-out dataset $\mathcal{D}_{\text{val}}$ throughout valuation.

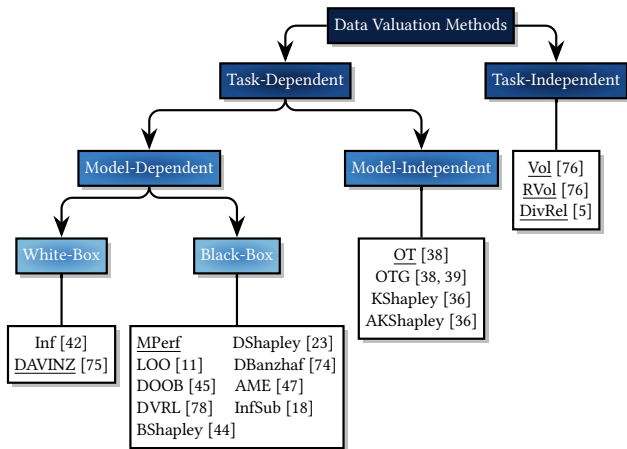


Figure 2: A taxonomy for data valuation methods (coarse-grained methods are underlined)

3.2 Coarse-Grained Data Valuation Methods

We describe coarse-grained data valuation methods, which target a dataset $\mathcal{D}$ and take as input the context C , consisting of an ML task $\mathcal{T}$, a model $\mathcal{M}$ trained on $\mathcal{D}$, and a hold-out set $\mathcal{D}_{\text{val}} = (z_j^v)_{j \in [n_{\text{val}}]}$. Performance is measured by $V(\mathcal{D}, \mathcal{M}, \mathcal{D}_{\text{val}})$, denoted by $V(\cdot)$ when only the training set varies. The context $C \in \mathcal{C}$ used by each method is made explicit below.

Model Performance (MPerf) is the baseline that equates the value of a dataset $\mathcal{D}$ with model performance: $v(\mathcal{D} \mid C) = V(\mathcal{D}, \mathcal{M}, \mathcal{D}_{\text{val}})$ with $C = (\mathcal{D}_{\text{val}}, \mathcal{M}_{\text{black}})$. Since this process is computationally prohibitive, SotA methods presented below typically develop efficient approximations of V that do not require training $\mathcal{M}$ on $\mathcal{D}$.

Optimal Transport (OT) [38] uses the Wasserstein distance W_c to compute the minimum cost to transform the empirical distribution of $\mathcal{D}$ into that of $\mathcal{D}_{\text{val}}$. Specifically, let μ and μ_{val} denote respectively the empirical distributions of $\mathcal{D}$ and $\mathcal{D}_{\text{val}}$. Then, the data value is $v(\mathcal{D} \mid C) = -W_c(\mu, \mu_{\text{val}})$ with $C = \mathcal{D}_{\text{val}}$. The metric on $\mathcal{Z}$ used to define W_c is a hybrid combining feature-space geometry with label information (the feature weight [4, 38] is fixed to 1 and the label weight λ_y is a parameter). The Wasserstein distance was proposed as an upper bound on the validation loss achievable by any model, hence as a proxy for V . This approximation was developed within the classification scope and remains valid when the model has sufficient capacity to achieve low training loss.

DAVINZ [75] approximates V for deep neural networks by means of the Neural Tangent Kernel (NTK) [34], requiring white-box access to $\mathcal{M}$ ’s parameters and gradients, to approximate the goodness of fit of $\mathcal{M}$ on $\mathcal{D}$ at initialization. Specifically, let $\Theta_0 = \Theta_0(\mathcal{D}, \mathcal{M})$ denote the Gram matrix of the model’s gradients evaluated at initialization at each data point in $\mathcal{D}$, and define $\text{NTK}(\mathcal{D}, \mathcal{M}_{\text{white}}) := (\hat{Y}\Theta_0^{-1}\hat{Y})^{1/2}$, with $\hat{Y}$ the residual of the label vector Y . The distribution shift between $\mathcal{D}$ and $\mathcal{D}_{\text{val}}$ is measured by the Maximum Mean Discrepancy (MMD) [25, 61] resulting in combined value: $v(\mathcal{D} \mid C) = -\lambda \cdot \text{NTK}(\mathcal{D}, \mathcal{M}_{\text{white}}) - \text{MMD}(\mathcal{D}, \mathcal{D}_{\text{val}})$ with $C = (\mathcal{D}_{\text{val}}, \mathcal{M}_{\text{white}})$ and λ is set to balance the two terms across datasets per [75]. The matrix Θ_0 is approximated via the Diagonal Block Approximation [75] with n_{batch} number of blocks, this parameter thus controlling the approximation granularity.

Robust Volume (RVol) [76] evaluates $\mathcal{D}$ via a geometric measure of diversity [1, 63, 66]. A baseline (Vol), defines the value as $v(\mathcal{D}) = \sqrt{\det(X^T X)}$ ($C = \emptyset$), but is sensitive to duplicate samples since replication can inflate the volume without increasing true diversity. To address this, RVol compresses the data by partitioning the feature space into n' cubes of equal side length ω and aggregating the data points within each cube (e.g., by their mean), producing a compressed matrix $\tilde{X} \in \mathbb{R}^{n' \times d}$. Then, given parameter $\alpha \in (0, 1)$, the data value is:

$$v(\mathcal{D}) = \sqrt{\det(\tilde{X}^T \tilde{X})} \prod_{i=1}^{n'} \frac{\alpha^{\psi_i+1} - 1}{\alpha - 1},$$

where ψ_i is the number of samples in cube $i \in [n']$. Per [76], $\alpha = 1/(n'\beta)$ where β represents a control parameter for a robustness guarantee.

DivRel [5] is a task-agnostic data valuation based on a divergence–relevance measure between the target dataset $\mathcal{D}$ and an external reference dataset $\mathcal{D}_{\text{ref}}$. It applies to situations where a user holds preliminary data and seeks to enrich it, prior to defining any concrete downstream objective. Let $X \in \mathbb{R}^{n \times d}$ and $X_{\text{ref}} \in \mathbb{R}^{n' \times d}$ denote respectively the feature matrices of $\mathcal{D}$ and $\mathcal{D}_{\text{ref}}$, and let $(e_i)_{i \in [d]}$ denote the eigenvectors of $\frac{1}{n} X_{\text{ref}}^\top X_{\text{ref}}$ with corresponding eigenvalues $(\lambda_i)_{i \in [d]}$. The variance of dataset $\mathcal{D}$ along direction e_i is given by $\hat{\lambda}_i = \frac{1}{n} e_i^\top X^\top X e_i$. Diversity and relevance are defined as $\text{Div} = \left(\prod_{i=1}^d \frac{|\lambda_i - \hat{\lambda}_i|}{\max\{\lambda_i, \hat{\lambda}_i\}} \right)^{1/d}$, $\text{Rel} = \left(\prod_{i=1}^d \frac{\min\{\lambda_i, \hat{\lambda}_i\}}{\max\{\lambda_i, \hat{\lambda}_i\}} \right)^{1/d}$, respectively, so that $\text{Div} + \text{Rel} \leq 1$. Then, the data valuation is the convex combination: $v(\mathcal{D} \mid \mathcal{D}_{\text{ref}}) = \alpha \text{Div} + (1 - \alpha) \text{Rel}$ with $\alpha \in [0, 1]$ controlling the divergence–relevance trade-off. The method is task-agnostic and privacy-preserving (only principal directions are shared), but requires selecting α and assumes that $\mathcal{D}$ and $\mathcal{D}_{\text{ref}}$ share the same feature space.

3.3 Fine-Grained Data Valuation Methods

We now present attribution operators (Def. 5), assigning fine-grained values to a family of data units $\mathcal{U} = (U_i)_{i \in [m]}$. Note that LOO, DShapley and DBanzhaf can work with any data valuation v , in which case their context is directly inherited from v , while the remaining fine-grained methods are specifically designed for MPerf: $v = V(\cdot)$.

Leave-One-Out (LOO) [11, 42] defines the fine-grained value of U_i as its *marginal contribution*: $\phi_i = v(\mathcal{D} \mid C) - v(\mathcal{D} \setminus U_i \mid C)$, $i \in [m]$. Computing all m fine-grained values thus requires making $(m + 1)$ calls to v . When the coarse value function is model performance $V(\cdot)$ the cost becomes impractical for complex models (e.g., neural networks). LOO is most useful when each U_i is sufficiently large, as a very small set of points rarely contributes significantly to model performance. **Shapley Value** [62], from cooperative game theory, fairly distributes a game’s total payoff among players and provides a theoretical foundation for data valuation. Each unit U_i is treated as a player, with value defined as the weighted average of its marginal contributions across all subsets of $\mathcal{U}$:

$$\phi_i = \sum_{S \subseteq [m] \setminus \{i\}} w(S, m) [v(\mathcal{D}_{S \cup \{i\}} \mid C) - v(\mathcal{D}_S \mid C)], \quad (1)$$

where $\mathcal{D}_S := \bigcup_{j \in S} U_j$ and $w_{\text{Shapley}}(S, m) := \frac{|S|!(m - |S| - 1)!}{m!}$ is the probability that U_i appears at position $|S| + 1$ in a uniform random permutation of $\mathcal{U}$. The Shapley value is the unique attribution operator satisfying: *Symmetry* (equivalent contributors receive equal value), *Zero Contribution* (non-influential units receive zero), *Efficiency* (values sum to $v(\mathcal{D} \mid C)$), and *Additivity* (values are additive over v). We describe four key works using the Shapley value for data valuation, addressing its quality or computational efficiency.

In what follows, the v used is $V(\cdot)$. Context is defined as $C = (\mathcal{D}_{\text{val}}, \mathcal{M}_{\text{black}})$, except for DOOB, which does not require $\mathcal{D}_{\text{val}}$ and OTG which uses $v = \text{OT}$; $C = \mathcal{D}_{\text{val}}$.

Data Shapley (DShapley) [23] adapts the Shapley value [62] to the practical setting of ML data valuation by using model performance as the coarse-grained value function. However, the exact computation requires $O(2^m)$ model trainings, which is computationally infeasible. The Shapley value (Eq. 1) can equivalently be written as an expectation over a uniformly random permutation π of the data units: $\phi_i = \mathbb{E}_{\pi \sim \Pi} V(U_i^\pi \cup \{U_i\}) - V(U_i^\pi)$, where U_i^π denotes the set of data units preceding U_i in permutation π . The contribution of DShapley is a Monte Carlo estimator of this expectation from T randomly sampled permutations of the data units, thereby reducing the computational cost to $O(Tm)$ model retrains. A truncation heuristic further applies early stopping within each permutation whenever marginal contributions fall below a threshold, exploiting the observation that data units added late contribute negligibly to model performance. Despite these optimizations, DShapley inherits the fixed symmetric weighting of the Shapley value, motivating subsequent extensions.

Beta Shapley (BShapley) [44] generalizes the Shapley value by replacing the symmetric weighting with a Beta-distribution-based weighting scheme: $w_{\text{beta}}(S, m) \propto \frac{B(\beta + |S|, m - |S| + \alpha - 1)}{B(\alpha, \beta)}$,

where $B(a, b)$ is the Beta function. Different choices of (α, β) shift the emphasis toward sets S with smaller, larger, or intermediate sizes. Setting $\alpha = \beta = 1$ recovers the standard Shapley weighting.

The methods described below are designed for the pointwise setting, i.e., $U_i = z_i$ and $m = n$.

Data Banzhaf (DBanzhaf) [74] adapts Data Shapley by replacing the Shapley weighting with uniform weights independent of the size: $w_{\text{banzhaf}}(S, n) = \frac{1}{2^n - 1}$ for each $S \subseteq [n]$. This contribution improves robustness by reducing sensitivity to noise in $V(\cdot)$. To estimate ϕ_i efficiently, the authors propose the *Maximum Sample Reuse* (MSR) estimator: draw a sample $\mathcal{S}$ of random subsets i.i.d. from $\text{Unif}(2^{[n]})$. For each $i \in [n]$, partition $\mathcal{S}$ into $\mathcal{S}_{\ni i} = \{S \in \mathcal{S} : i \in S\}$ and $\mathcal{S}_{\not\ni i} = \mathcal{S} \setminus \mathcal{S}_{\ni i}$, and estimate $\phi_i \approx \frac{1}{|\mathcal{S}_{\ni i}|} \sum_{S \in \mathcal{S}_{\ni i}} V(\mathcal{D}_S) - \frac{1}{|\mathcal{S}_{\not\ni i}|} \sum_{S \in \mathcal{S}_{\not\ni i}} V(\mathcal{D}_S)$. This reduces computational complexity, as it is only requires $|\mathcal{S}|$ evaluations of $V(\cdot)$, reused across all units.

Influence Subsample (InfSub) [18] exploits Influence Functions [42] for data valuation. It estimates influence using a subsampling approach. The value formulation is identical to DBanzhaf, except that the random sample $\mathcal{S}$ is drawn i.i.d. from $\mathcal{S}(M) := \{S \subseteq [n] : |S| = M\}$. In other words, instead of sampling subsets of arbitrary sizes, InfSub focuses on more informative subsets using a fixed size M , or equivalently a proportion $p = M/n$, that can be tuned.

KNN-Shapley (KShapley) [36] addresses the computational bottleneck of DShapley by restricting $V(\cdot)$ to a K -nearest neighbor (kNN) classifier, where the value of a subset U with respect to a validation point (x^v, y^v) is the fraction of its kNNs with correct labels. This structure admits a recursive formula, processing training points in a single pass sorted by distance to x^v from furthest to nearest: $\phi(z_{\alpha_n} \mid x^v, y^v) = \frac{1[y_{\alpha_n} = y^v]}{n}$, and for $i \in [n - 1]$, $\phi(z_{\alpha_i} \mid x^v, y^v) = \phi(z_{\alpha_{i+1}} \mid$

$x^v, y^v) + \frac{1[y_{\alpha_i} = y^v] - 1[y_{\alpha_i+1} = y^v]}{K \cdot \min\{K, i\}}$, where α_i is the index of the i -th NN to x^v . The final value is averaged over $\mathcal{D}_{\text{val}}$: $\phi_i = \frac{1}{n_{\text{val}}} \sum_{j=1}^{n_{\text{val}}} \phi(z_i | z_j^v)$, reducing complexity from $O(2^n)$ to $O(n \log n)$ per validation point with no model retraining. For large n and n_{val} , the authors propose an LSH-based approximation [36], AKShapley, using the 2-stable LSH variant [13] for Euclidean distance. Its parameters are ϵ , defining an expanded neighborhood $K^* = \max(K, \lceil 1/\epsilon \rceil)$; L , the number of hash tables; and α , the number of hash bits per table. The bucket width t and normalization distance d are auto-tuned per [36]. Both KShapley and AKShapley are model-independent.

Influence Functions (Inf) [42] The context is $C = (\mathcal{D}, \mathcal{D}_{\text{val}}, \mathcal{M}_{\text{white}})$. Influence Functions are a classical tool from robust statistics [11, 30, 32] that measure the effect of upweighting a data point on model parameters and predictions; in the context of data valuation, they are used to approximate the leave-one-out (LOO) value for differentiable models without retraining. Model parameters are obtained via empirical risk minimization (ERM) [70]: $\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(z_i, \theta)$. To analyze the effect of a data point $z_i \in \mathcal{D}$, consider the perturbed objective where z_i is upweighted by ϵ : $\hat{\theta}_{\epsilon, z_i} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(z_i, \theta) + \epsilon \mathcal{L}(z_i, \theta)$. The influence on the parameters is $\mathcal{I}_{\text{param}}(z_i) = \left. \frac{d\hat{\theta}_{\epsilon, z_i}}{d\epsilon} \right|_{\epsilon=0} = -H_{\hat{\theta}}^{-1} \nabla_{\theta} \mathcal{L}(z_i, \hat{\theta})$, and the influence on the loss on z^v is

$$\mathcal{I}_{\text{loss}}(z_i, z^v) = \nabla_{\theta} \mathcal{L}(z^v, \hat{\theta})^{\top} \mathcal{I}_{\text{param}}(z_i)$$

where $z^v \in \mathcal{D}_{\text{val}}$ is a validation point. Removing z_i corresponds to setting $\epsilon = -\frac{1}{n}$, so the LOO effect on the validation loss is approximated as $-\frac{1}{n} \mathcal{I}_{\text{loss}}(z_i, z^v)$. The fine-grained data value is $\phi_i \approx -\frac{1}{n} \cdot \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{z^v \in \mathcal{D}_{\text{val}}} \nabla_{\theta} \mathcal{L}(z^v, \hat{\theta})^{\top} H_{\hat{\theta}}^{-1} \nabla_{\theta} \mathcal{L}(z_i, \hat{\theta})$, where $H_{\hat{\theta}} = \frac{1}{n} \sum_{i=1}^n \nabla_{\theta}^2 \mathcal{L}(z_i, \hat{\theta})$ is the Hessian of the training loss. While Influence Functions do not require retraining the model for each data point, they remain computationally expensive as Hessian computation requires $O(np^2 + p^3)$, where p is the number of model parameters. Scalability has been addressed via Hessian-vector products (HVP) and Conjugate Gradient (CG) techniques [3, 42]. A number of methods have been proposed for measuring influence using various proxies [18, 51, 55], extend to more complex models [10, 27], and enable the valuation of non-singleton data units [41].

Data-OOB (DOOB) [45] exploits the bagging framework, avoiding the need for a holdout set entirely. Given B bootstrap datasets sampled with replacement from $\mathcal{D}$ at proportion p , a weak learner $\hat{f}_b$ is trained on each sample $\mathcal{D}_b$, $b \in [B]$. The value of each data point z_i is then estimated via out-of-bag (OOB) evaluations, z_i serving as an independent validation data point for all the bootstrap samples that do not include it: $\phi_i := \frac{\sum_{b=1}^B 1[z_i \notin \mathcal{D}_b] T(y_i, \hat{f}_b(x_i))}{\sum_{b=1}^B 1[z_i \notin \mathcal{D}_b]}$, where $T(y_i, \hat{f}_b(x_i))$ measures the goodness of fit of $\hat{f}_b$ on z_i (e.g., correctness in classification or negative squared error in regression). The value of z_i is thus the average score across all bootstrap models for which it was out-of-bag. The complexity is $O(B)$, as the B weak learners are

trained once and reused across all data points, making it more computationally efficient than the Shapley-based methods.

Average Marginal Effect (AME) [47] AME formulates data valuation by treating the inclusion of each point z_i as a random treatment. The value ϕ_i is defined as the average marginal effect: $\phi_i := \mathbb{E}_S [V(S \cup \{z_i\}) - V(S)]$, where the expectation is with respect to a random choice of S sampled from a user-defined distribution on the power set $\mathcal{P}(\mathcal{D})$. Lin et al. [47] showed that AME can be efficiently estimated via a LASSO regression. Specifically, for a regularization parameter $\lambda > 0$, a predefined transformation function $g : \{0, 1\}^n \rightarrow \mathbb{R}^n$, and a collection of subsets $S \subseteq \mathcal{P}(\mathcal{D})$, the estimate $\hat{y}_i = \phi_i$ is obtained by solving: $\min_{\mathbf{y} \in \mathbb{R}^n} \frac{1}{|S|} \sum_{S \in S} (V(S) - g(\mathbf{1}_S)^{\top} \mathbf{y})^2 + \lambda \sum_{i=1}^n |y_i|$, where $\mathbf{1}_S \in \{0, 1\}^n$ is the indicator vector of S . The parameter λ is autotuned via Lasso cross-validation per [47]. $\mathcal{P}(\mathcal{D})$ samples from proportions $\{0.2, 0.4, 0.6, 0.8\}$ per [47], resulting in $|S|$ subset models trained per proportion. A regression then jointly estimates each data point's value, improving estimation under sparsity assumptions.

Data Valuation using Reinforcement Learning (DVRL) [78]

frames data valuation as a bilevel optimization problem, where a data value estimator $h : \mathcal{Z} \rightarrow [0, 1]$ assigns importance weights to training points in the loss function of a predictor f_{θ} . Specifically, h solves $\min_h \sum_{j=1}^{n_{\text{val}}} \mathcal{L}(y_j^v, f_{\theta}(x_j^v))$ s.t. $\theta^h = \arg \min_{\theta} \sum_{i=1}^n h(x_i, y_i) \cdot \mathcal{L}(y_i, f_{\theta}(x_i))$. The outer objective minimizes validation loss, while the inner objective trains f_{θ} h -weighted samples. Since sample selection involves a discrete Bernoulli draw $s_i \sim \text{Ber}(h(x_i, y_i))$, which is non-differentiable, h is optimized via reinforcement learning over a number of RL epochs with batch size B , using validation performance as a reward signal to up-weight valuable points and down-weight noisy ones. The data value for each training point is given by $\phi_i = h(x_i, y_i) \in [0, 1]$ for $i \in [n]$.

OT Gradient (OTG) [38, 39] extends OT coarse valuation to pointwise granularity by defining the value of each data point z_i as the sensitivity of W_c to perturbations in its probability mass: $\phi_i := -\frac{\partial W_c(\mu, \mu_{\text{val}})}{\partial \mu(z_i)}$. LAVA [38] follows an exact approach where the full OT problem is solved, and gradients are obtained as a byproduct, requiring the construction of a cost matrix with memory complexity $O(n \times n_{\text{val}})$, which is prohibitive. SAVA [39] addresses this by decomposing the main problem into smaller subproblems processed using a batch approach, the batch size thus being a tunable hyperparameter.

4 Experimental Evaluation

4.1 Environment

Experiments were conducted on the UM6P cluster [40]. Coarse-grained methods were ran on an NVIDIA A100-SXM4 (80 GB) GPU with dual AMD EPYC 7713 CPUs, allocated 32 cores and 250 GB RAM per job via SLURM (CUDA 12.8). Fine-grained methods were ran on dual Intel® Xeon® Platinum 8276L CPUs (2.2R0 GHz, 56 cores, 192 GB RAM). The GPU was required to compute MPerf on complex models; and was thus used across all coarse-grained methods for consistency. All baselines were implemented using Python 3.9 and were executed using a

single thread to ensure a fair comparison, as not all implementations in the framework readily support multithreading.

4.2 Experimental Setup

Methods. We evaluate four coarse-grained methods: OT [4, 38], Vol [76], RVol [76], DAVINZ [75], and the baseline MPerf. We also cover eleven fine-grained methods: LOO [23], DShapley [23] (using truncated Monte Carlo per [37]), DBanzhaf [74], KShapley/AKShapley [36], DOOB [45], InfSub [18] (representing influence-based methods for non-differentiable models [37]), AME [47], DVRL [78], OTG (LAVA/SAVA) [38, 39], and a Random (Rand) baseline. All fine-grained methods use MPerf as the underlying data valuation, except OTG which uses OT; OT values are reported as negated outputs. All methods use their most efficient publicly available implementations: OpenDataVal [37] for fine-grained methods, [75] for Vol/RVol and DAVINZ, and the original implementations for AKShapley [36], LAVA [38] and SAVA [39]. DivRel was excluded as it values a target dataset relative to a reference dataset other than $\mathcal{D}_{val}$, making it incompatible with our experimental setting without crafting a dedicated scenario for this method alone.

Table 1: Datasets, Models & Tasks (C: Classification with number of classes, R: Regression, -: Benchmark)

Dataset	Domain	$ \mathcal{Z} $	$ \mathcal{D} $	d	Model	$\mathcal{T}$
<i>Image</i>						
DogFish [42]	-	1,800	1,500	2,048	LogReg	CI (2)
MNIST [46]	-	60,000	10,000	784	CNN	CI (10)
CIFAR10 [43]	-	50,000	40,000	3,072	ResNet-18/LogReg	CI (10)
ImageNet [14]	-	50,000	50,000	12,288	ResNet-50/ResNet-18	CI (100)
HAM [48]	Medicine	700	700	2,048	-	CI (7)
WebSkin	Medicine	2,800	2,800	2,048	LogReg	CI (7)
<i>Tabular</i>						
Adult [15]	Social Science	48,842	30,000	14	LogReg	CI (2)
Connect4 [69]	Gaming	67,557	40,000	42	LogReg	CI (3)
CoverType [8]	Environment	581,012	10,000	54	MLP2	CI (7)
Hep [7]	Physics	10,500,000	7,000,000	28	LogReg	CI (2)
CASP [57]	Biology	45,730	10,000	9	MLP10	R

Datasets. We include eleven datasets (Tab. 1) covering classification and regression across image and tabular modalities, spanning diverse domains including medicine, biology, physics, social science, gaming, and environment, with sizes from 700 to $\sim 7\text{M}$, dimensionality from 14 to 12,288, and up to 100 classes. Our experimental design reflects a deliberate trade-off between breadth and scale. The core evaluation uses datasets of 1K–10K points, consistent with the literature [23, 37, 44, 45, 74], where these sizes are considered challenging for data valuation and enable fair comparison across all methods. Methods requiring black-box model access retrain the model multiple times (some, e.g., DShapley and DBanzhaf, with exponential complexity) and have only been evaluated up to 1K points in prior work [23, 37, 45, 74]. To address scalability, we complement the core evaluation with a dedicated study on larger datasets (up to 7M points), where only tractable methods are included. To our knowledge, the only prior data valuation study at million scale covered only KShapley [36], evaluated

with a single validation point, whereas our scalability study includes KShapley, AKShapley, DOOB, DVRL, LAVA, and SAVA. We use the original dataset name for the largest subset, with a size suffix for smaller subsets (e.g., Hep-1K, CIFAR10-10K).

Scenarios. Classification over homogeneous tabular and image datasets has been the dominant setting in the data valuation literature [23, 37, 38, 45]. Our study follows this convention while also extending the evaluation to regression, which is supported by a subset of the coarse-grained methods and remains underexplored in existing benchmarks. Methods are evaluated on efficiency, accuracy, and memory footprint, varying datasets, dataset sizes and data types. The best methods are then evaluated on scalability. Since coarse-grained and fine-grained methods differ in the granularity of the valuation output, each family is evaluated on distinct scenarios. Coarse-grained methods are assessed on four scenarios: 1) correlation to MPerf; 2) robustness to increasing feature/label noise (to evaluate if the value scores drop with increasing noise); 3) invariance to replication (to evaluate if the score values remain stable with increasing replication); and 4) sensitivity to data diversity (to evaluate if the value scores increase with greater coverage of the data domain [63, 76]). Fine-grained methods are assessed across three scenarios: 1) low-value detection (covering the impact of low-value removal on model performance and noisy data detection since the noisy ground truths are known); 2) high-value detection (covering only high-value removal since the high-value ground truths are not known as a point that was not noised may still be low-value); and 3) domain adaptation (which combines both low-value and high-value detection simultaneously). Since the majority of fine-grained data valuation methods operate at the pointwise level, we defer groupwise valuation to future work and focus our evaluation on the pointwise setting, which provides the broadest and most meaningful comparison across methods.

Measures. We use the following measures: *Efficiency* is measured by wall-clock time; *model performance* by accuracy (classification) and negative MSE (regression); *model performance correlation* by Pearson correlation (r) [38, 75]; *noise robustness* and *diversity sensitivity* by Spearman correlation (ρ) [64]; *replication invariance* by the Coefficient of Variation (CV) [52]; *low-value detection* by F1-score, precision, and recall, where predicted noisy points are identified as the lower-mean cluster from 2-means clustering on data values [37, 44, 45, 74]; *high-value detection* by PDR@k (Performance Drop after Removal at k), the drop in model performance after removing the top $(k \times 100)\%$ highest-valued points (we adopt PDR@0.2 [37] as it balances sensitivity and interpretability, avoiding negligible changes at small k and dataset-size confounding at large k); and *domain adaptation* by model accuracy.

Procedure. Datasets are preprocessed consistently across all methods, including encoding, splitting, and noise injection. Three disjoint subsets are sampled from each domain $\mathcal{Z}$: the target dataset $\mathcal{D}$, the validation set $\mathcal{D}_{val}$ ($|\mathcal{D}_{val}| = 20\%|\mathcal{D}|$), and the test set $\mathcal{D}_{test}$ ($|\mathcal{D}_{test}| = 50\%|\mathcal{D}|$), used only for evaluation. Predefined splits (DogFish, MNIST, CIFAR10, ImageNet, Hep) exclude $\mathcal{D}_{test}$ from re-splitting. In domain adaptation, $\mathcal{D}$ is the full source domain while $\mathcal{D}_{val}$ and $\mathcal{D}_{test}$ are drawn

from the target distribution. For tabular data, categorical features are one-hot encoded and numerical features are z-score standardized. For image data, features are extracted using pre-trained models (ResNet50 [31] for ImageNet, ResNet18 [31] for CIFAR10, CNN [46] for MNIST, InceptionV3 [65] for HAM and WebSkin), yielding dimensions of 2048, 512, 1568, and 2048 respectively; DogFish is provided in embedding form. For pointwise scenarios, we inject 20% label noise (classification) [23, 37, 44, 45, 74, 78], keeping $\mathcal{D}_{\text{val}}$ clean [23, 38, 78]. For coarse-grained scenarios, label and feature noise levels range from 2% to 100%; diversity is assessed via nested sampling [1, 63, 66]; and replication factors range from $1 \times$ to $10 \times$ [2, 38, 76]. Each method is tuned across its parameter space to identify its Pareto-optimal accuracy/efficiency configuration. Each valuation run is repeated 10 times with a 24-hour time budget; results are averaged over 6 runs (discarding the two highest and two lowest) and reported with 95% confidence intervals.

4.3 Methods Parameter Tuning

This section studies hyperparameter tuning for the valuation methods. For RVol, OT, LAVA, and KShapley, whose parameters do not affect computational cost, we simply select the values that maximize accuracy. For the remaining methods, accuracy typically increases with computational cost before plateauing (and sometimes declining). We therefore choose the lowest-cost configuration that attains maximal accuracy, corresponding to the elbow points in Figures 4-5, where the Pareto frontier is the increasing portion of each curve. To the best of our knowledge, this is the most comprehensive parameter sensitivity analysis of data valuation methods published to date. Due to space constraints, we only include a representative subset of parametrization results to illustrate key trends; the complete results are provided in [12] (Figs. 4-5).

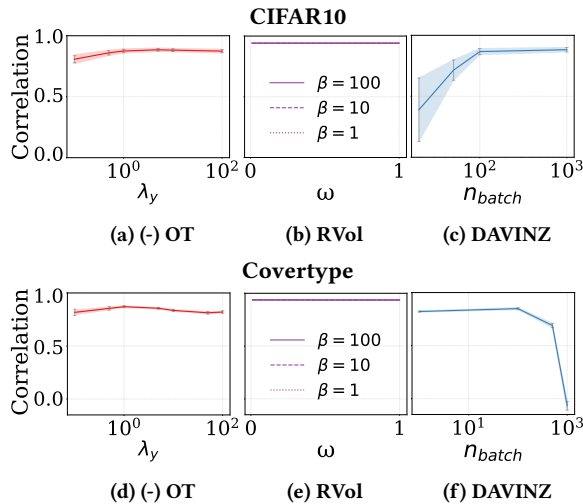


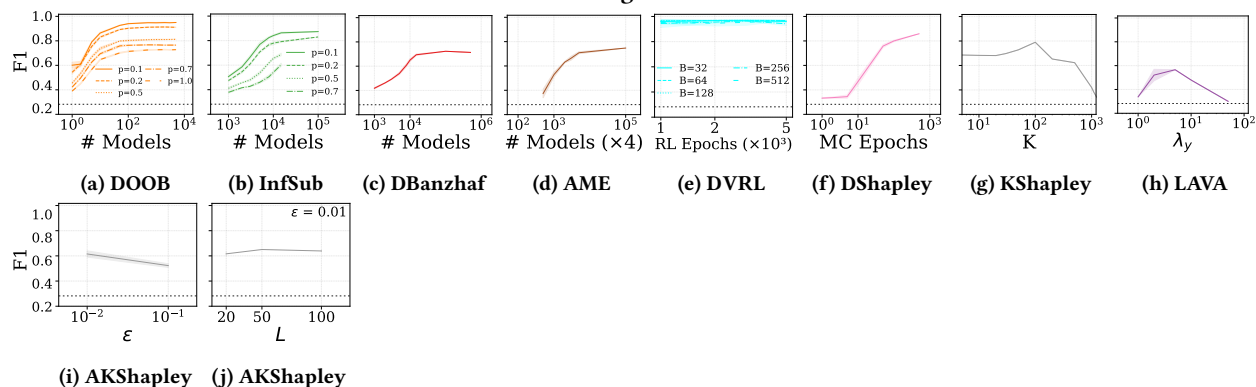
Figure 3: Coarse-grained methods parameter tuning

Coarse-Grained Methods. We tune coarse-grained methods using model performance correlation (r) on two modalities:

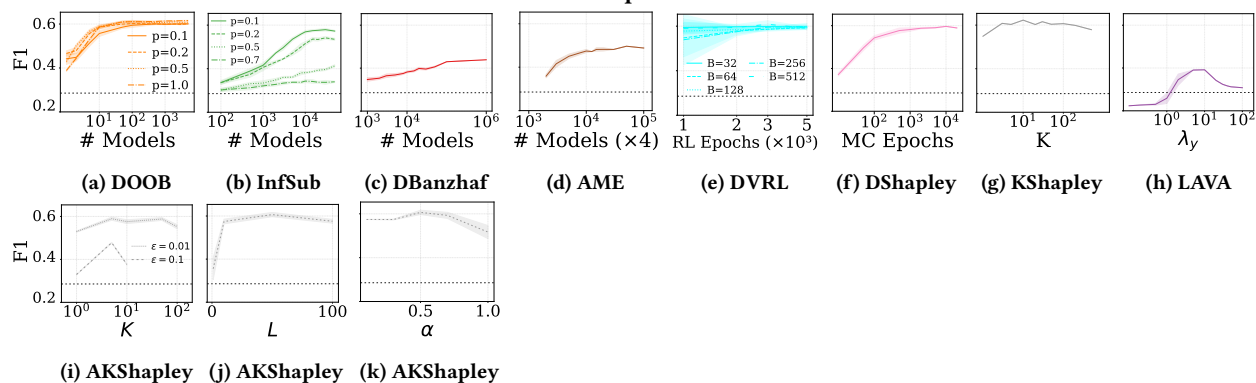
Covertime (tabular, MLP) and CIFAR10 (image, ResNet-18). For each, we generate 100 bootstrap samples of sizes 1K–10K, train to convergence, and use validation performance as ground truth, a setup that accommodates OT and DAVINZ, proposed for classification and deep neural networks respectively. We additionally evaluate on the CASP regression task (MLP), where OT is excluded as it was proposed for classification. Since CASP and Covertime are both tabular datasets, and the same model architecture (MLPs) was used to solve their associated learning task, the same tuning procedure applies directly. Figure 3 shows that OT achieves the best correlation at $\lambda_y = 1$ on both datasets, which we fix for all experiments. Vol requires no tuning, and RVol’s correlation remains stable across all configurations of ω and α , confirming that its replication robustness component does not degrade Vol’s effectiveness; we fix $\omega = 0.1$ and $\beta = 10$ following [76]. For DAVINZ, $n_{\text{batch}} = 1$ (exact NTK) is optimal for Covertime (Fig. 3f), as the MLP makes exact NTK computation feasible. For CIFAR10, $n_{\text{batch}} = 100$ provides the best accuracy/efficiency trade-off (Fig. 3c); we adopt this value for all complex architectures (CIFAR10, MNIST, ImageNet) per [75].

Fine-Grained Methods. We tune each method across three scenarios (low-/high-value detection and domain adaptation) on all datasets, using Hep-1K results to summarize key trends; Hep-1M parameters are reused for Hep. Full tuning results are in (Figs. 4-5). DOOB converges with lower latency using a small number of models (100 for $|\mathcal{D}| \leq 1K$, 10 otherwise), with smaller proportions improving efficiency while maintaining performance. InfSub performs best in the low-value scenario with a fixed subset size of ~ 100 points (e.g., $p = 0.1$ at 1K, $p = 0.01$ at 10K), as small subsets amplify the detectability of noisy points. High-value and domain adaptation scenarios require larger subsets to reliably include high-value points while preserving contrast; at 1K, both regimes coincide at $p = 0.1$, but diverge at larger scales (e.g., at 100K: $p = 0.001$ for low-value, $p = 0.1$ for high-value). A useful heuristic for InfSub, DBanzhaf, and AME is at least $10 \times$ the dataset size for low-value detection and $100 \times$ for high-value detection. DVRL [78] benefits from smaller batch sizes (32: tabular, 128: image) with 3,000–5,000 RL epochs at small-to-medium scales (1K–100K), while a batch size of 512 improves performance and convergence at larger scales (1M–7M). DShapley [23] uses 1,000 Monte Carlo iterations as a reasonable accuracy/efficiency trade-off. KShapley [36] uses $K = 10$ for low-value detection across most datasets and higher K for high value detection (e.g., $K = 500$ for Hep-1K). For AKShapley [36], we set ϵ to 10^{-2} , 10^{-3} , and 10^{-4} for small, medium (Hep-10K, CIFAR-10), and large datasets ($\geq 100K$) respectively. We set $L = 20$ for datasets $\leq 10K$ and $L = 100$ beyond, with $\alpha = 0.5$ throughout. LAVA [38] uses label weights of 10 (tabular) and 5 (image), and remains feasible up to 100K samples (requiring over 931GB at 1M). LAVA and SAVA are equivalent when the OT cost matrix fits in memory; therefore we tune SAVA’s [39] batch size only on Hep-1M, obtaining a setting of 1,024, and apply it also to Hep. The full results are in [12] (Figs. 4-5)).

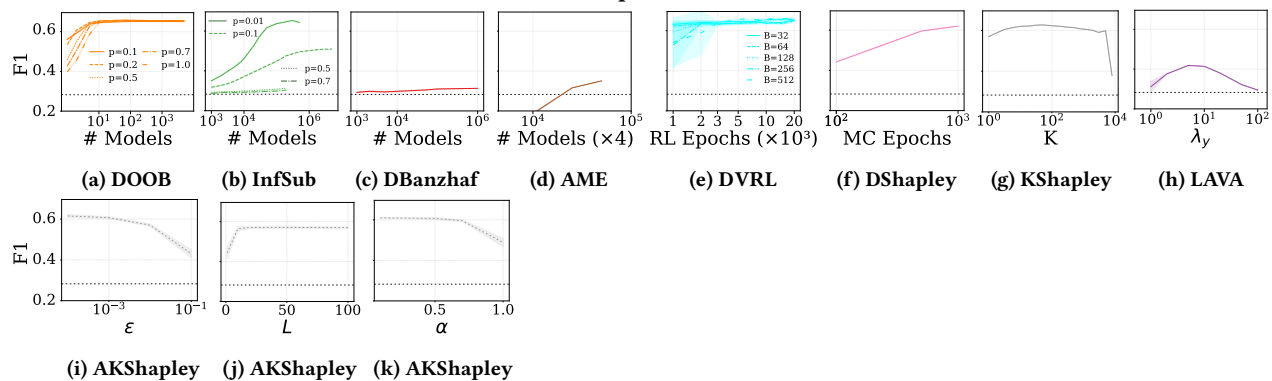
DogFish



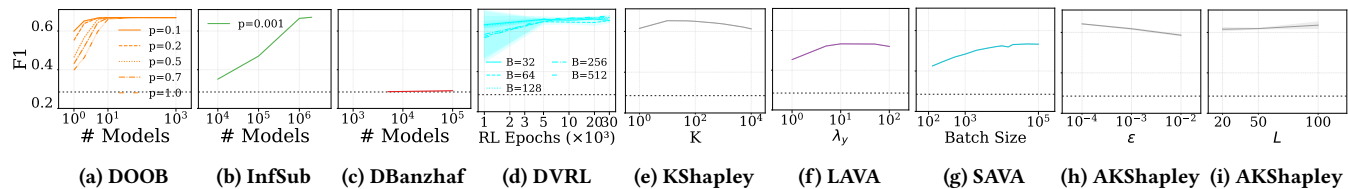
Hep-1K



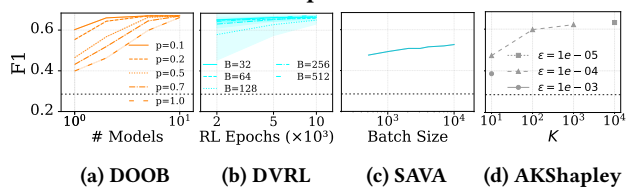
Hep-10K



Hep-100K



Hep-1M



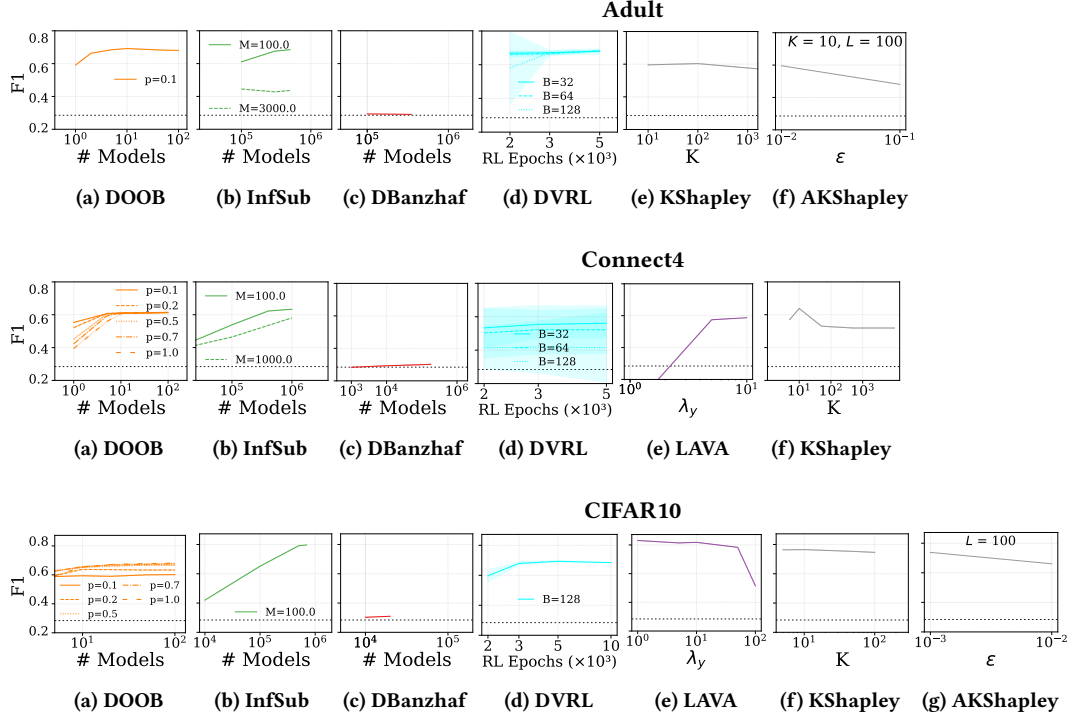


Figure 4: Fine-grained methods parameter tuning (low-value detection, dotted line represents Rand)

4.4 Coarse-Grained Methods Evaluation

We evaluate OT, DAVINZ, Vol, and RVol on CoverType, MNIST, CIFAR10, and ImageNet across four dimensions: model performance prediction, noise robustness, diversity sensitivity, and replication invariance. We focus on classification, as OT was specifically designed for this task, and include Vol alongside RVol since the latter is its replication-aware extension. We additionally evaluate model performance prediction on the CASP regression dataset using a 10-layer MLP, excluding OT as it does not support it. Valuation covers the full ImageNet domain and up to 10K samples for all other datasets.

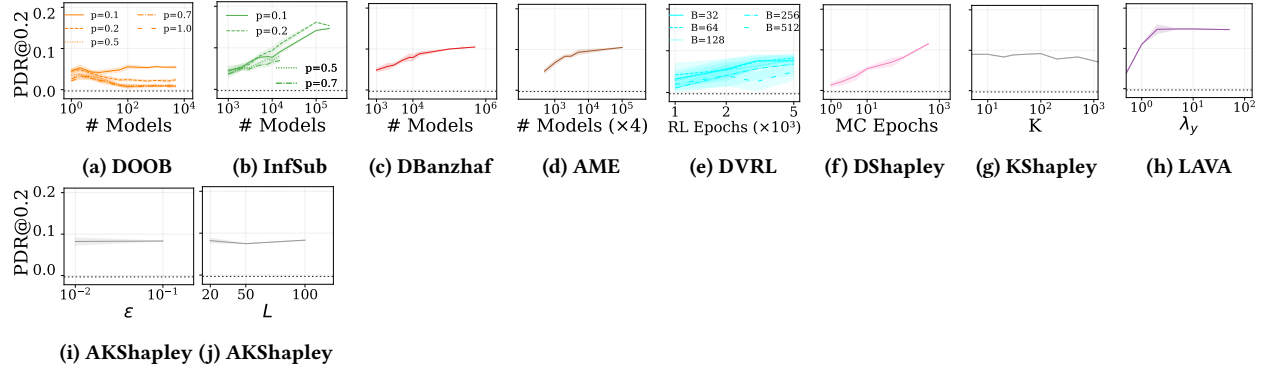
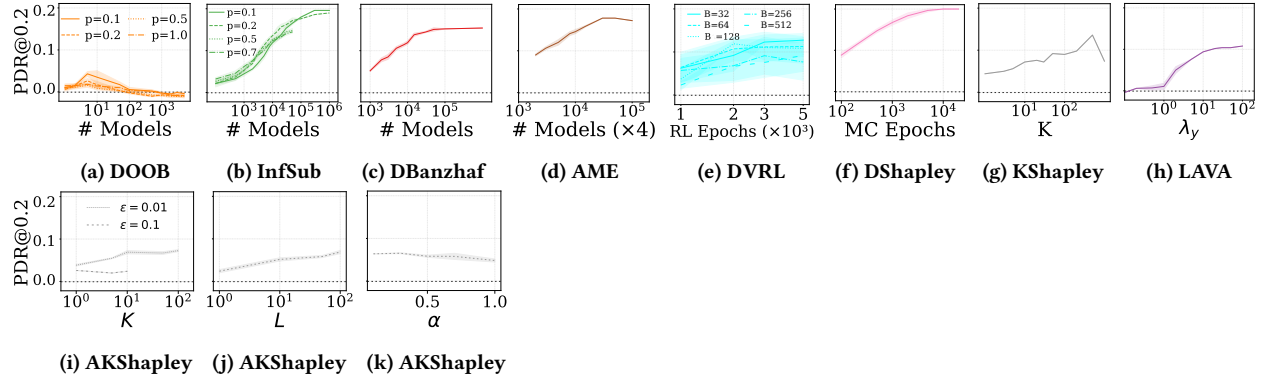
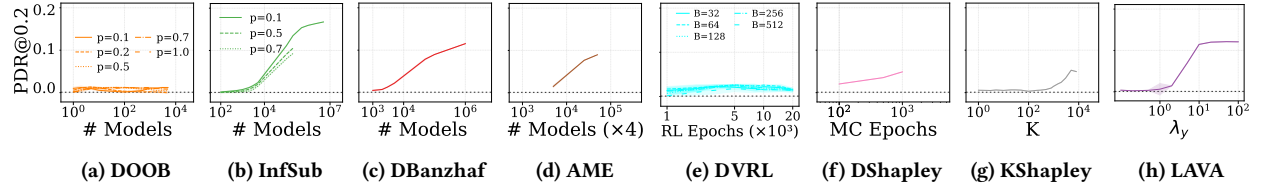
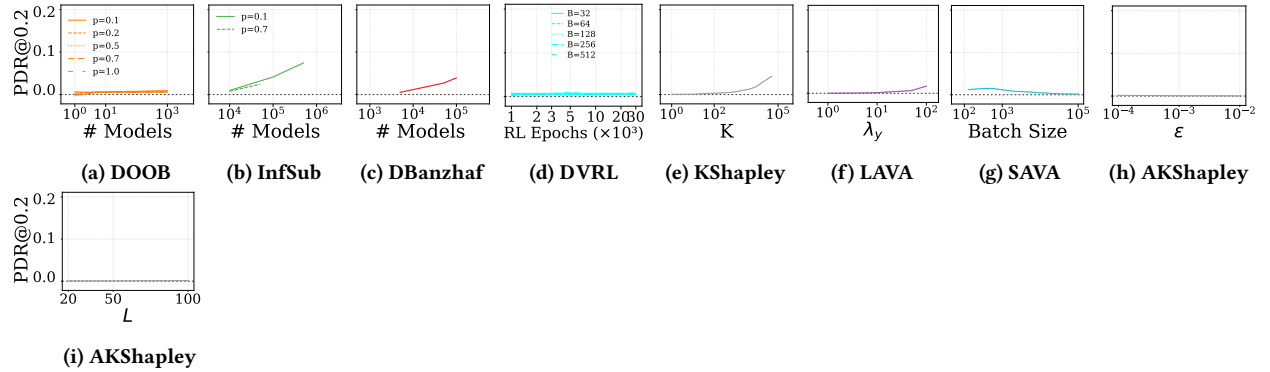
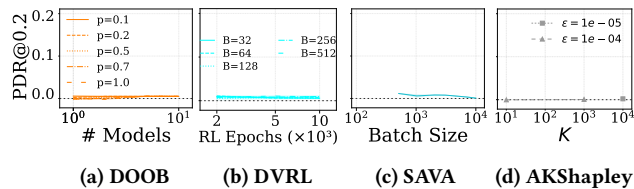
Model Performance Prediction. For each dataset, we train models on random bootstrap subsets of $\mathcal{D}$ and measure Pearson correlation r between the estimated coarse-grained value and validation performance on $\mathcal{D}_{\text{val}}$. For DAVINZ, we additionally report its MMD and NTK components separately. Results are in Table 2 (ImageNet omitted as MPerf ground truth computations exceed the 24-hour threshold). All methods perform best on MNIST, whose classes are well separated [26] and whose task is simpler than Covertype and CIFAR10 (accuracy: 94%–98% vs. 60%–70% on Covertype and 40%–70% on CIFAR10). Vol and RVol achieve the best overall performance, as dataset volume directly captures data diversity and coverage, a reliable proxy for model performance in this clean setting. RVol reduces to Vol here since all bootstraps exhibit similar replication. OT lags on Covertype and CIFAR10, suggesting it is a less reliable performance proxy on complex datasets.

For DAVINZ, NTK consistently outperforms MMD, as MMD measures distributional distance in raw feature space, which is uninformative on complex image datasets where pixel statistics lack semantic meaning. All methods except DAVINZ use pretrained embeddings [36–38], which provide more meaningful representations for measuring data similarity. DAVINZ operates on raw data by design, as its authors argue embeddings introduce valuation bias [75], which may explain its degraded performance on CIFAR10. Unlike MMD, OT also incorporates label information, explaining its advantage over MMD. NTK’s performance degrades on CASP relative to Covertype despite using MLPs for both datasets, likely due to the task difference (regression vs. classification) and deeper architecture (10 vs. 2 layers), worsening the NTK. DAVINZ’s correlation is further reduced by the NTK–MMD weighting term.

Table 2: Model performance prediction (r)

Dataset	(-) OT	Vol/RVol	(-) MMD	(-) NTK	DAVINZ
Covertype	0.87 ± 0.01	0.94 ± 0.00	0.71 ± 0.00	0.96 ± 0.00	0.82 ± 0.00
MNIST	0.97 ± 0.00	0.97 ± 0.00	0.67 ± 0.00	0.97 ± 0.00	0.93 ± 0.01
CIFAR10	0.88 ± 0.01	0.94 ± 0.00	0.16 ± 0.00	0.87 ± 0.02	0.86 ± 0.02
CASP	–	0.97 ± 0.00	0.53 ± 0.00	0.87 ± 0.01	0.56 ± 0.00

Noise Robustness. Table 3 reports Spearman correlation ρ between value score rankings (decreasing order) and noise level rankings (increasing order), where higher ρ indicates greater robustness. Under label noise, OT is the only robust

DogFish**Hep-1K****Hep-10K****Hep-100K****Hep-1M**

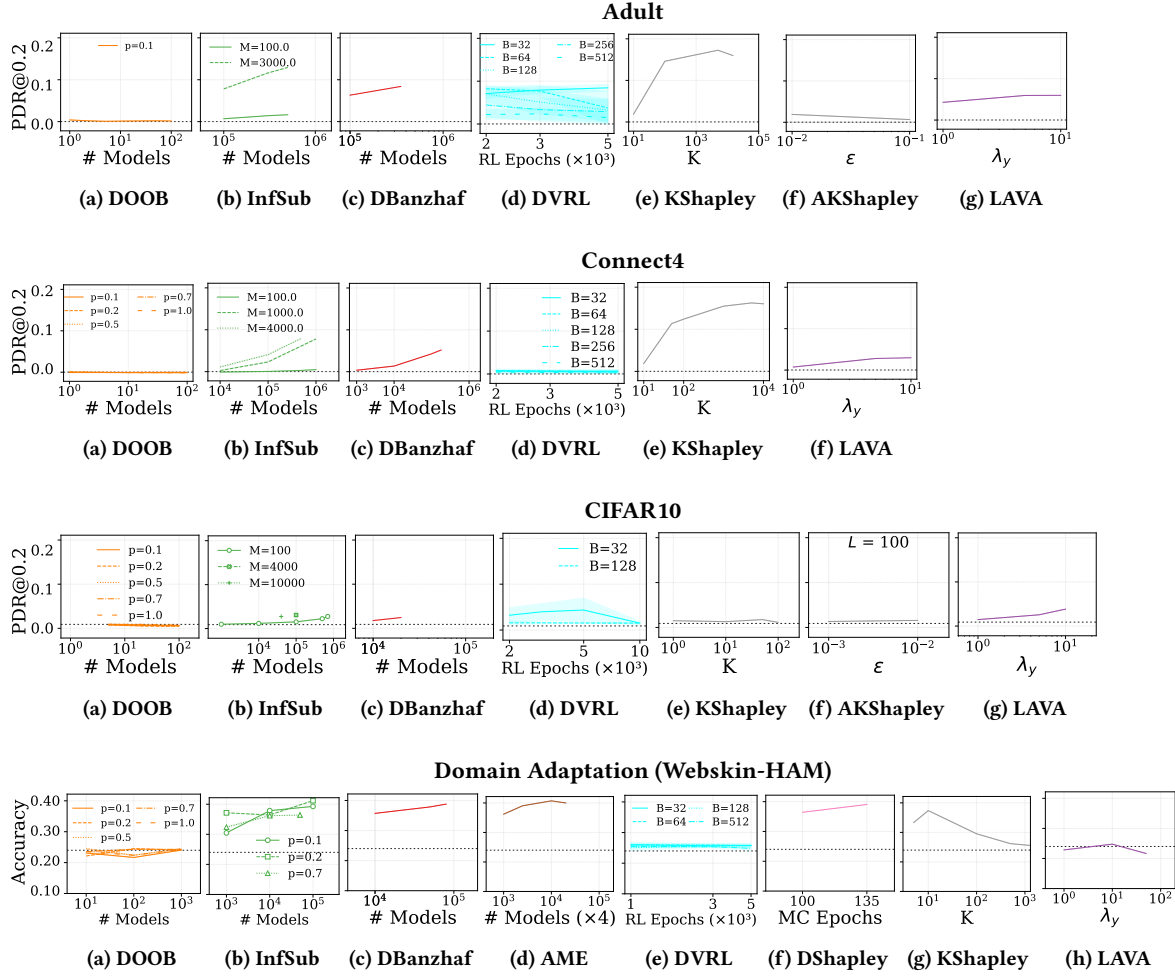


Figure 5: Fine-grained methods parameter tuning (high-value detection and domain adaptation, dotted line represents Rand for high-value detection and baseline accuracy for domain adaptation)

method ($\rho \approx 1$): it measures transport distance jointly over features and labels and is inherently sensitive to outliers [38], treating label noise as such [21]. Vol, RVol, and MMD completely fail as they operate purely on feature distributions. NTK shows partial but unreliable robustness, and DAVINZ inherits this instability since MMD contributes nothing to label noise detection. Under feature noise, OT and MMD both achieve $\rho \approx 1$: feature corruption increases distributional distance from $\mathcal{D}_{\text{val}}$, which both methods correctly detect. Vol and RVol exhibit the opposite behavior ($\rho \approx -1$): feature noise artificially inflates the data volume, causing noisy datasets to appear more diverse and receive incorrectly higher scores. A similar behavior occurs with NTK, where it shows the inverse pattern, as feature noise disrupts the gradient structure and regularizes the NTK kernel. However, DAVINZ is able to recover under feature noise by leveraging its MMD component, achieving consistently positive correlations.

Table 3: Noise robustness ρ (N/A means undefined)

Dataset	(-) OT	Vol/RVol	(-) MMD	(-) NTK	DAVINZ
<i>Label Noise</i>					
Coverttype	0.97 ± 0.01	N/A	N/A	0.31 ± 0.26	0.31 ± 0.26
MNIST	0.98 ± 0.00	N/A	N/A	0.78 ± 0.56	0.78 ± 0.56
CIFAR10	0.99 ± 0.00	N/A	N/A	0.95 ± 0.03	0.95 ± 0.03
ImageNet	1.00 ± 0.00	N/A	N/A	0.05 ± 0.62	0.05 ± 0.62
<i>Feature Noise</i>					
Coverttype	0.98 ± 0.01	-0.97 ± 0.01	0.97 ± 0.01	-0.26 ± 0.38	0.68 ± 0.27
MNIST	1.00 ± 0.00	-0.99 ± 0.00	0.99 ± 0.00	-1.00 ± 0.00	0.96 ± 0.01
CIFAR10	1.00 ± 0.00	-1.00 ± 0.00	0.99 ± 0.00	-1.00 ± 0.00	0.97 ± 0.01
ImageNet	0.98 ± 0.00	-0.78 ± 0.02	0.98 ± 0.01	-0.99 ± 0.00	0.82 ± 0.02

Diversity Sensitivity. Table 4 reports ρ between valuation scores and dataset size, where higher ρ indicates greater diversity sensitivity. Vol and RVol achieve perfect correlation, as volume grows monotonically with diversity by design [76]. OT, MMD, and NTK perform well on most datasets but struggle

on ImageNet, likely due to its scale and class count. OT fails completely ($\rho = -1$): with $|\mathcal{D}_{val}|$ fixed at 5K, the growing dataset increasingly deviates from it, which OT misinterprets as decreasing value. MMD shows high variance on ImageNet, possibly due to high-dimensional pixel space computations. NTK also struggles on ImageNet, suggesting reduced reliability in high dimensions. DAVINZ inherits OT’s failure. Overall, Vol and RVol are the only methods sensitive to diversity, while OT and MMD conflate diversity with proximity to the validation distribution, which is misleading when the validation set is small or unrepresentative, causing them to undervalue large, diverse datasets that deviate from it.

Table 4: Diversity sensitivity (ρ)

Dataset	(-) OT	Vol/RVol	(-) MMD	(-) NTK	DAVINZ
Coverttype	0.84 ± 0.20	1.00 ± 0.00	0.89 ± 0.15	0.98 ± 0.01	0.98 ± 0.01
MNIST	1.00 ± 0.00	1.00 ± 0.00	0.91 ± 0.09	1.00 ± 0.00	0.98 ± 0.03
CIFAR10	0.99 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	1.00 ± 0.00	0.99 ± 0.00
ImageNet	-1.00 ± 0.00	1.00 ± 0.00	0.62 ± 0.42	-1.00 ± 0.00	-0.86 ± 0.18

Table 5: Replication Invariance (CV)

Dataset	(-) OT	Vol	RVol	(-) MMD	(-) NTK	DAVINZ
Coverttype	0.01 ± 0.00	0.10 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.09	0.25 ± 0.05
MNIST	0.05 ± 0.00	0.57 ± 0.00	0.07 ± 0.00	0.00 ± 0.00	0.20 ± 0.03	0.10 ± 0.01
CIFAR10	0.07 ± 0.01	0.15 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.06 ± 0.01	0.03 ± 0.00
ImageNet	0.00 ± 0.00	0.09 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.14 ± 0.01	0.07 ± 0.00

Replication Invariance. Table 5 reports the coefficient of variation (CV) of valuation scores under increasing replication: a lower CV indicating greater invariance. MMD achieves zero CV across all datasets, as empirical distributions are unaffected by duplication [25]. OT is largely invariant, with a slightly higher CV on MNIST (0.05) and CIFAR10 (0.07), likely due to numerical sensitivity. Vol exhibits high CV, particularly on MNIST ($CV = 0.57$), as geometric volume grows directly with duplication [76], a known limitation that RVol was designed to address. Indeed, RVol achieves zero CV in most cases, except on MNIST ($CV = 0.07$) perhaps due to numerical instabilities in volume computation. NTK shows poor invariance, particularly on Coverttype ($CV = 0.50$): replication introduces duplicate samples, making the NTK matrix ill-conditioned and destabilizing the NTK term which relies on matrix inversion [75]; this is amplified on Coverttype where $n_{batch} = 1$ (exact NTK) is used. DAVINZ is moderately affected by NTK’s instability. Overall, MMD and RVol are the most replication-invariant methods, OT is invariant with minor numerical instability, while Vol and NTK are the least invariant.

4.5 Fine-Grained Methods Evaluation

Recall that fine-grained methods attribute the value of $\mathcal{D}$, estimated using a coarse-grained method, to its individual points. All methods discussed in this paper use MPerf, except LAVA and SAVA which use OT. Plots exclude baselines that could not complete the valuation process within 24 hours.

High-Value Data Detection. Figure 6 reports high-value detection results, where effective methods exhibit a sharp and early accuracy drop as high-value points are progressively removed. InfSub consistently stands out, exhibiting an approximately linear decline in accuracy as data is removed across most datasets (Figs. 6a–6d, 6g–6h): by fixing the subset size, it uses compact, informative subsets where performance is stable and redundancy is low. AME and DBanzhaf also achieve strong early drops on datasets up to 10K and 100K respectively; AME’s earlier cutoff stems from the growing cost of its regression-based value estimation, while DBanzhaf’s simpler averaging approach scales further. KShapley performs moderately on smaller datasets and approaches optimal behavior on Adult and Connect4 (Figs. 6g, 6h), but degrades beyond 100K (Fig. 6d), likely due to its model-independence. Notably, its decrease is non-linear and sharper compared to InfSub and DBanzhaf, which decrease more uniformly (Figs. 6d, 6g, 6h). A uniform decrease is preferable, as it indicates more accurate high data value ranking. AKShapley performance degrades relative to KShapley as the dataset size grows, due to its approximate nature. DShapley is effective on Hep-1K and DogFish but cannot scale beyond 10K, as each Monte Carlo permutation requires model training proportional to dataset size. As dataset size grows from 1K to 7M (Figs. 6b–6f), the magnitude of the accuracy drop decreases across all methods, as removing the top 20% has diminishing relative impact and high-value points become harder to identify in larger, more uniform pools. DOOB, DVRL, and LOO show limited ability to identify high-value points, behaving close to random removal. LAVA and SAVA shows moderate performance across all datasets.

Low-Value Data Detection. We assess low-value detection via low-value removal (Fig. 7) and noisy data discovery (Fig. 8). For low-value removal, data points are ranked by value and removed from lowest to highest, with performance reported on $\mathcal{D}_{test}$. Methods exhibit three behaviors: accuracy improvement (harmful points removed), preservation (redundant points removed), or degradation (valuable points incorrectly ranked low). The first two are desirable. On DogFish (Fig. 7a), most methods improve accuracy, with DVRL achieving the strongest gains. On Hep-1K (Fig. 7b), DShapley, InfSub, KShapley, and AKShapley achieve clear improvement. As dataset size increases, improvement gives way to preservation, with InfSub, DOOB, and DVRL consistently maintaining stable performance even at larger scales (Figs. 7e–7f). DBanzhaf, LOO, AME, KShapley, AKShapley, LAVA, and SAVA exhibit unstable behavior, consistently degrading accuracy across multiple settings. For noisy data discovery, we report recall on the lowest-ranked points, where high recall at small inspection fractions is desired. DOOB and DVRL consistently achieve the best performance, with DVRL slightly surpassing DOOB. DShapley and InfSub follow closely with comparable performance despite their fundamental design difference: DShapley relies on exhaustive Monte Carlo sampling while InfSub achieves similar results through selective fixed-size sampling, highlighting that carefully designed sampling strategies can match exhaustive approaches at a fraction of the cost. KShapley and AKShapley achieves comparable performance through

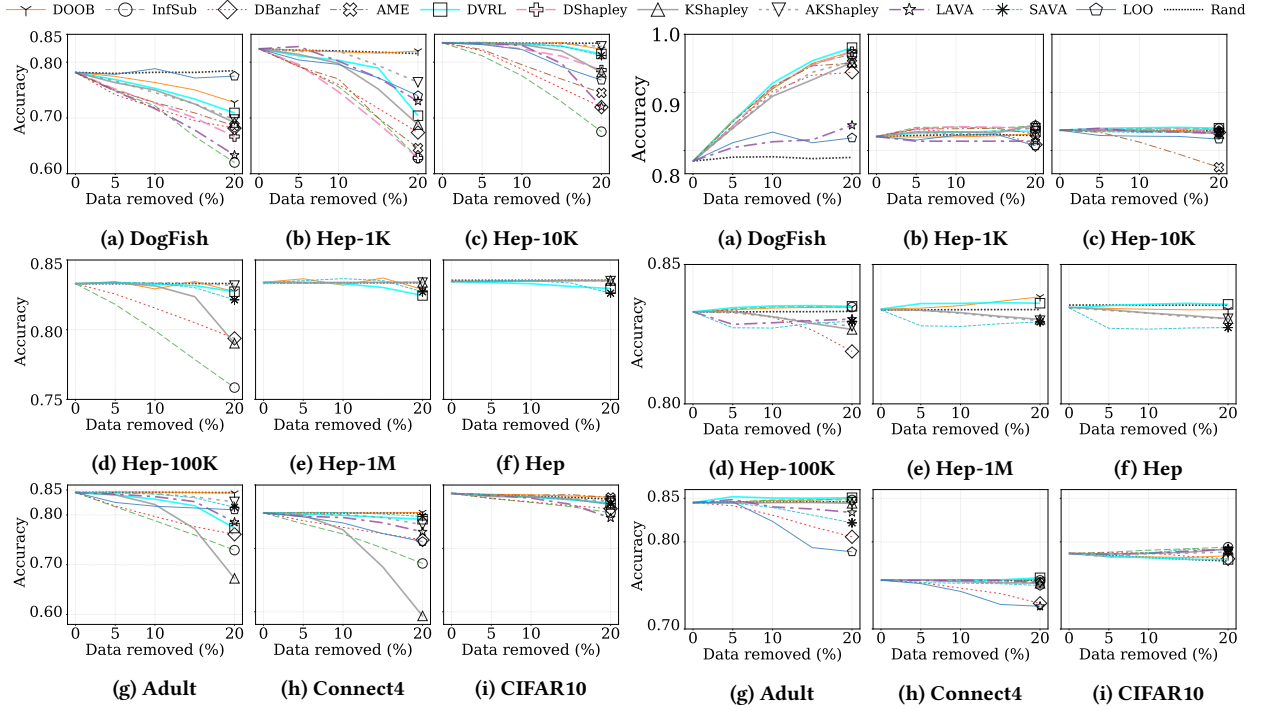


Figure 6: High-value removal

Figure 7: Low-value removal

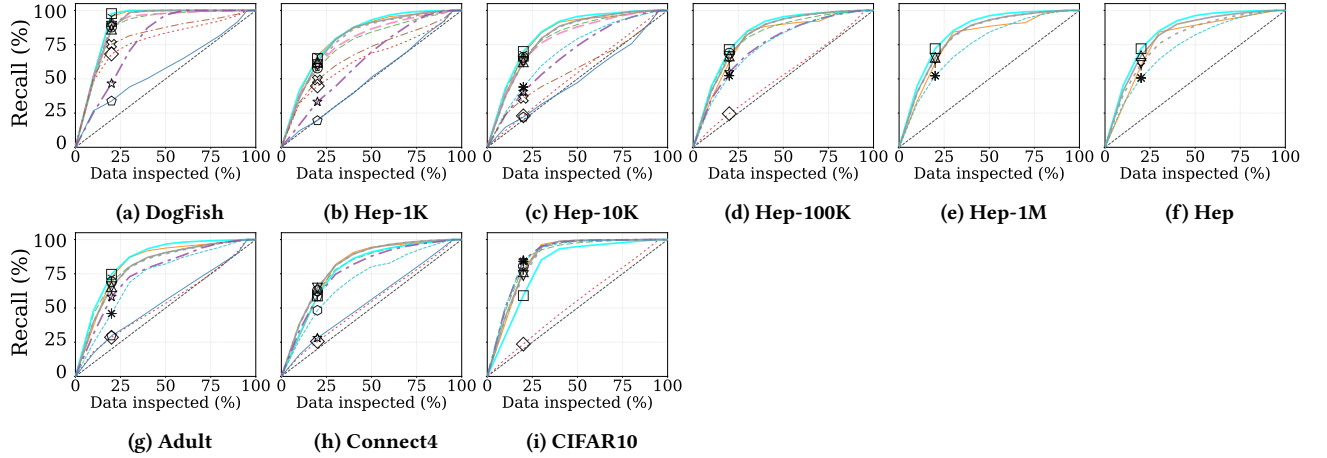


Figure 8: Noisy data detection

an entirely different mechanism, relying solely on data geometric properties without any model access. DBanzhaf and AME show moderate performance on small datasets but degrade to near-random on larger ones, as their uniform sampling becomes ineffective at scale. The contrast between InfSub and DBanzhaf, which share the same formulation differing only in sampling strategy, further underscores the importance of sampling. LAVA performs poorly on small datasets but improves consistently with scale (Figs. 8c–8d), as larger datasets provide a more representative validation set, improving the

reliability of its OT-based value estimates [38], and reaching near-optimal behavior on Connect4 (Fig. 8h) and CIFAR10 (Fig. 8i). In the fine-grained setting, each dataset is an independent valuation scenario with its own validation context, which explains why $\mathcal{D}_{val}$ grows with the dataset, unlike the coarse-grained setting where a fixed $\mathcal{D}_{val}$ is shared across all target datasets. LAVA and SAVA were originally evaluated on images [38], and our experiments confirm strong performance on CIFAR10. However, both underperform on tabular data. Notably, while [37] reported near-zero performance for LAVA,

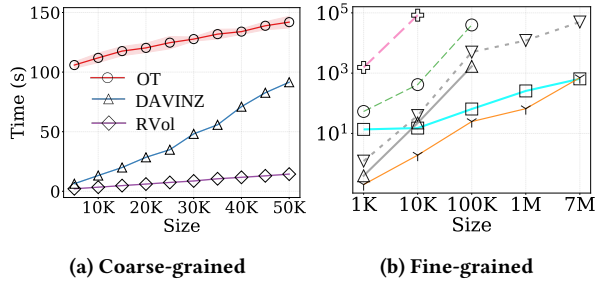


Figure 9: Scalability

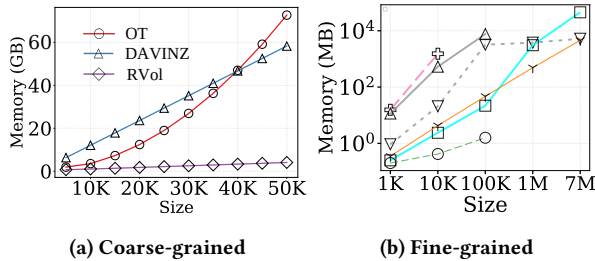


Figure 10: Memory Footprint

our results show that proper parameter tuning recovers competitive performance, underscoring the importance of careful parametrization. Finally, in line with prior findings [23], LOO behaves close to Rand.

Domain Adaptation. The goal is to improve model performance on the target domain HAM by exploiting data valuation on the source domain WebSkin. The target domain is split into 280 validation ($\mathcal{D}_{\text{val}}$) and 420 test ($\mathcal{D}_{\text{test}}$) samples using a stratified split. Each method assigns value scores to WebSkin images, used as sample weights during training; the resulting model is evaluated on $\mathcal{D}_{\text{test}}$, and improvement over the unweighted baseline (24% accuracy) reflects each method’s ability to identify target-relevant data. This scenario is more challenging than previous ones: it involves two distributions, a high-noise regime from uncurated web data, and requires methods to simultaneously detect low-value samples and reliably rank high-value ones. AME and InfSub achieve the best performance ($\sim 41\%$, a $1.7\times$ improvement) within ~ 13 hours, followed by DBanzhaf and DShapley at 39% within 24 hours. KShapley reaches within 3% of the best accuracy in seconds, owing to its model-free nature. The strong performance of these methods can be attributed to their direct use of $\mathcal{D}_{\text{val}}$ to define data value: by measuring each training point’s contribution to validation performance, they naturally align with the domain adaptation objective of identifying target-relevant data. In contrast, DVRL shows only marginal improvement despite ~ 17 minutes of runtime; LAVA and DOOB behave similarly to the baseline; and LOO performs worse than the baseline as its values are dominantly zero. Methods that do not directly optimize value with respect to $\mathcal{D}_{\text{val}}$ performance fail to capture what matters in this cross-domain setting.

4.6 Scalability

We evaluate scalability by studying accuracy/efficiency/memory trade-offs under increasing dataset size.

Efficiency/Accuracy. For coarse-grained methods, we analyze time efficiency on ImageNet (50K samples, 100 classes), as accuracy has already been discussed (§4.4). Figure 10a shows that RVol runs in nearly constant time, while OT and DAVINZ scale linearly with dataset size, with OT being more computationally expensive. For fine-grained methods, we use mislabeled data detection on Hep subsets, as label noise is more challenging to detect than feature noise [21] and provides ground truths for evaluation. We measure wall-clock time for each method to reach a target F1 score ($58\% \pm 2\%$ at 1K, $63\% \pm 2\%$ at 10K, $66\% \pm 1\%$ at 100K–7M), reporting the earliest time the F1 enters the tolerance window. On Hep-1M and Hep, the validation set is reduced to 10K to include KShapley. As shown in Fig. 10b, DOOB and DVRL scale to the 7M dataset. From 1K to 100K, AKShapley’s approximate search is slower than KShapley; however, beyond 100K, AKShapley extends where KShapley fails due to memory constraints, though with a lower F1 (60% vs. 66% at 1M, 57% vs. 68% at 7M compared to DOOB/DVRL). InfSub scales to 100K and DShapley to 10K, constrained by exhaustive retraining. LAVA and SAVA are excluded as they do not reach the target F1. Among model-dependent methods, exhaustive training (DShapley, InfSub) limits scalability, while reinforcement learning (DVRL) and bagging (DOOB) scale better. KShapley and AKShapley are model-independent, yet remain costly, as computing introduces a looping cost and a dependency validation set size.

Footprint. For coarse-grained methods, memory is the main overhead. DAVINZ grows linearly, reaching ~ 60 GB at 50K. OT grows quadratically, approaching the 80 GB GPU memory limit at 50K, while RVol reaches only 4 GB at 50K. For fine-grained methods, DShapley and KShapley incur the highest memory footprint below 100K: DShapley due to storing permutations, and KShapley due to storing large data batches to compute pairwise distance computations. AKShapley is more memory-efficient than KShapley. DVRL and DOOB follow a similar linear memory trend, remaining in the few MB range, while InfSub has the smallest memory footprint overall. Beyond 100K, DOOB remains the most memory-efficient at scale alongside AKShapley, which stays nearly constant at 1M and 7M. DVRL requires more memory at 1M and 7M due to larger RL batches. Full memory results are in [12] (see Fig.10).

5 Lessons Learned & Research Directions

In this section, we summarize the key lessons learned and the research directions they motivate.

Parameter Sensitivity and Tuning. InfSub performs best only after careful tuning; the optimal sampling proportion (Fig. 4b), set to yield a subset of 100 points for low value and sampling proportion $p = 0.1$ for high value, differs significantly from the original paper [18] where $p = 0.7$ is recommended. This is likely due to differences in experimental setup: InfSub was originally designed for memorization analysis [18]

and later adopted for data valuation [37], while our setup targets low/high-value detection. For DOOB, although the original paper does not emphasize the sampling proportion [45], our experiments show it critically affects performance: smaller proportions reduce computational cost while preserving accuracy on tabular data, and improve both accuracy and efficiency on image data [12]. These findings expose a gap in the literature: parameter sensitivity is often overlooked and tuning guidelines are rarely provided. Future work should establish *principled tuning recommendations* across tasks and datasets.

Scalability. Most fine-grained methods do not scale beyond small datasets, and those that do (DOOB, DVRL and both exact and approximate KShapley) are largely limited to mislabeled data detection (Figs. 8f and 6f). The number of model trainings is a key bottleneck. DOOB requires fewer models (5-100) than reported in the literature (800-1000) [37, 45], improving efficiency without sacrificing accuracy. This effect is amplified at larger scales where even fewer models are sufficient. In contrast, InfSub, AME, and DBanzhaf require at least $10 \times$ more models than reported [37, 45], making them prohibitive at scale. Coarse-grained methods such as RVol scale but fail under noise, motivating *noise-aware volume measures*. KShapley stands out by avoiding repeated model training altogether, scaling naturally to large datasets and providing a strong baseline, highlighting the relevance of nearest-neighbor structures for data valuation and opening new directions at the intersection with the *similarity search* community, including theoretical guarantees [17] for model-independent methods.

High-Value Detection and Its Limits. Removing the top 20% of points produces almost no accuracy change at large scale (Figs. 6e and 6f), suggesting either that high-value identification degrades with dataset size or that existing methods are insufficient in this regime. This asymmetry with low-value detection stems from a fundamental difference in ground truth availability: noise is an intrinsic point-level property providing a clear low-value proxy (Fig. 8 and Tab. 3), whereas high value is assessed only through point removal, which is difficult to interpret consistently across methods. Besides, value scores alone cannot reliably distinguish high from low value across methods operating on different scales. This calls for *realistic benchmarks* with ground-truth high- and low-value subsets and *interpretable valuation scores* that clarify whether scores carry absolute meaning or serve solely as relative rankings.

Towards Groupwise Valuation. A deep limitation of pointwise valuation is that it can fail to identify the most valuable *group*: a point may appear low-value in isolation yet contribute significantly within a specific subset, and vice versa. This motivates methods that natively support *adaptive aggregation* (Def. 6) and satisfy *group-consistency* (Def. 7), going beyond pointwise coverings to directly value overlapping subsets, a largely unexplored research direction.

Broader Task and Data Support. The data valuation literature has predominantly focused on classification over homogeneous tabular and image datasets [23, 36–38, 45], and our study follows this convention. Real-world ML pipelines, however, involve a much richer variety of tasks and data types. Methods

should be extended to retrieval, clustering, ranking, object detection, and LLM fine-tuning, each requiring different notions of model performance and data value. On the data side, current methods assume homogeneous feature spaces, whereas real-world data is often heterogeneous. Future work should develop *multimodal valuation* methods and extend benchmarking to graphs, time series, and text.

Privacy, Validation, and Legal Perspectives. Most methods require full access to $\mathcal{D}$ and $\mathcal{M}$, which is rarely feasible due to privacy constraints and legal regulations, motivating *privacy-preserving valuation* compliant with data and AI regulations. Besides, relying on a shared validation set introduces additional fragility: its quality, size, and class balance directly affect valuation outcomes [35, 76], and distribution shift between validation and deployment data undermines stability across domains (§ 4.5). Moreover, a provider with prior knowledge of the validation set could strategically craft contributions to appear more valuable than they are, undermining valuation integrity. Building robust defenses against such manipulation is a prerequisite for trustworthy data markets.

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